

## MAUNA LOA 5 NNE, HI, USA

WMO: 912950

Lat: 19.5354N

Lon: 155.5760W

Elev: 3407

StdP: 66.55

Time Zone: -10.00 (W10)

Period: 05-19

WBAN: 21514

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |            | Humidification DP/MCDB and HR |            |            |              |            |            | Coldest Month WS/MCDB |            |             |            | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|------------|-------------------------------|------------|------------|--------------|------------|------------|-----------------------|------------|-------------|------------|-----------------------|------------|--------------|
|               |             |            | 99.6%                         |            |            | 99%          |            |            | 0.4%                  |            | 1%          |            |                       |            |              |
|               | 99.6%       | 99%        | DP                            | HR         | MCDB       | DP           | HR         | MCDB       | WS                    | MCDB       | WS          | MCDB       | MCWS                  | PCWD       |              |
| (a)<br>2      | (b)<br>-0.8 | (c)<br>0.1 | (d)<br>-42.0                  | (e)<br>0.1 | (f)<br>8.5 | (g)<br>-38.1 | (h)<br>0.1 | (i)<br>8.5 | (j)<br>13.3           | (k)<br>4.9 | (l)<br>11.8 | (m)<br>6.1 | (n)<br>3.8            | (o)<br>N/A | (p)<br>0.688 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |            |             |            |             |            | Evaporation WB/MCDB |             |            |             |            |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|------------|-------------|------------|-------------|------------|---------------------|-------------|------------|-------------|------------|-------------|----------------------|------------|
|               |                        | 0.4%            |            | 1%          |            | 2%          |            | 0.4%                |             | 1%         |             | 2%         |             |                      |            |
|               |                        | DB              | MCWB       | DB          | MCWB       | DB          | MCWB       | WB                  | MCDB        | WB         | MCDB        | WB         | MCDB        | MCWS                 | PCWD       |
| (a)<br>6      | (b)<br>8.2             | (c)<br>16.6     | (d)<br>3.3 | (e)<br>15.6 | (f)<br>3.3 | (g)<br>14.9 | (h)<br>3.3 | (i)<br>9.0          | (j)<br>11.5 | (k)<br>8.4 | (l)<br>11.0 | (m)<br>8.0 | (n)<br>10.8 | (o)<br>5.6           | (p)<br>N/A |

| Dehumidification DP/MCDB and HR |             |            |            |            |            |            |            |            | Enthalpy/MCDB |             |             |             |             |             | Extreme<br>Max WB |
|---------------------------------|-------------|------------|------------|------------|------------|------------|------------|------------|---------------|-------------|-------------|-------------|-------------|-------------|-------------------|
| 0.4%                            |             |            | 1%         |            |            | 2%         |            |            | 0.4%          |             | 1%          |             | 2%          |             |                   |
| DP                              | HR          | MCDB       | DP         | HR         | MCDB       | DP         | HR         | MCDB       | Enth          | MCDB        | Enth        | MCDB        | Enth        | MCDB        |                   |
| (a)<br>7.9                      | (b)<br>10.1 | (c)<br>9.9 | (d)<br>7.2 | (e)<br>9.7 | (f)<br>9.6 | (g)<br>6.6 | (h)<br>9.3 | (i)<br>9.5 | (j)<br>36.2   | (k)<br>11.6 | (l)<br>34.7 | (m)<br>11.0 | (n)<br>33.5 | (o)<br>10.8 | (p)<br>10.6       |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 10.3              | 9.0  | 7.9 | -2.8                       | 18.6 | 1.5                | 0.6 | -3.9                                               | 19.0 | -4.8       | 19.4 | -5.6       | 19.7 | -6.7       | 20.2 |
|                   |      |     | -7.8                       | 10.0 | 0.9                | 0.5 | -8.5                                               | 10.3 | -9.0       | 10.6 | -9.5       | 10.9 | -10.2      | 11.2 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 7.7    | 6.3   | 5.4   | 5.9   | 7.0   | 8.2   | 9.5   | 9.3   | 9.3   | 9.0   | 8.6   | 7.1   | 6.4   |
|                                                                        | DBStd       | 2.48   | 2.56  | 2.47  | 2.19  | 2.11  | 2.12  | 1.89  | 1.88  | 1.68  | 1.60  | 1.63  | 1.92  | 2.20  |
|                                                                        | HDD10.0     | 917    | 116   | 129   | 128   | 93    | 63    | 31    | 35    | 34    | 38    | 49    | 88    | 112   |
|                                                                        | HDD18.3     | 3889   | 373   | 362   | 386   | 341   | 314   | 265   | 279   | 280   | 280   | 302   | 338   | 370   |
|                                                                        | CDD10.0     | 69     | 2     | 1     | 1     | 2     | 7     | 16    | 14    | 12    | 8     | 5     | 1     | 1     |
|                                                                        | CDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDH23.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 3.8    | 3.9   | 3.9   | 4.0   | 3.8   | 3.8   | 4.0   | 3.9   | 3.7   | 3.2   | 3.4   | 3.4   | 4.0   |
| (15) Precipitation                                                     | PrecAvg     | 776    | 91    | 57    | 79    | 47    | 45    | 36    | 59    | 51    | 57    | 52    | 70    | 65    |
|                                                                        | PrecMax     | 1367   | 368   | 156   | 202   | 204   | 188   | 114   | 154   | 96    | 151   | 120   | 295   | 174   |
|                                                                        | PrecMin     | 335    | 6     | 4     | 17    | 21    | 11    | 5     | 12    | 14    | 6     | 13    | 5     | 0     |
|                                                                        | PrecStd     | 259    | 85    | 44    | 62    | 38    | 40    | 22    | 39    | 21    | 38    | 30    | 68    | 64    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 15.4  | 14.5  | 14.8  | 15.4  | 17.0  | 17.4  | 17.7  | 17.3  | 16.6  | 16.2  | 14.4  | 14.8  |
|                                                                        |             | MCWB   | 2.1   | 1.2   | 2.5   | 2.4   | 3.4   | 4.0   | 3.1   | 3.6   | 3.0   | 2.5   | 2.2   | 1.2   |
|                                                                        | 2%          | DB     | 13.6  | 12.7  | 13.1  | 13.9  | 15.1  | 15.9  | 15.9  | 15.9  | 15.0  | 14.7  | 13.3  | 12.9  |
|                                                                        |             | MCWB   | 1.8   | 1.6   | 2.6   | 2.3   | 3.7   | 3.7   | 3.4   | 3.6   | 3.9   | 3.0   | 2.3   | 1.1   |
|                                                                        | 5%          | DB     | 12.3  | 11.3  | 11.8  | 12.7  | 13.9  | 14.9  | 14.9  | 14.7  | 13.9  | 13.6  | 12.2  | 11.8  |
|                                                                        |             | MCWB   | 1.3   | 1.5   | 2.4   | 2.9   | 3.9   | 3.7   | 3.5   | 3.7   | 4.0   | 3.5   | 2.5   | 1.2   |
|                                                                        | 10%         | DB     | 11.1  | 9.9   | 10.4  | 11.4  | 12.7  | 13.9  | 13.7  | 13.5  | 12.8  | 12.4  | 11.1  | 10.5  |
|                                                                        |             | MCWB   | 0.9   | 1.3   | 2.0   | 2.7   | 3.7   | 3.8   | 3.8   | 4.3   | 4.6   | 3.6   | 2.4   | 1.3   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 6.4   | 6.7   | 7.1   | 7.6   | 8.5   | 9.1   | 9.1   | 9.6   | 9.5   | 9.1   | 8.0   | 7.0   |
|                                                                        |             | MCDB   | 10.7  | 9.8   | 10.8  | 11.0  | 12.4  | 12.6  | 11.4  | 11.9  | 12.0  | 11.4  | 10.8  | 9.9   |
|                                                                        | 2%          | WB     | 5.2   | 5.6   | 5.9   | 6.8   | 7.6   | 8.1   | 8.2   | 8.8   | 8.7   | 8.3   | 7.0   | 5.8   |
|                                                                        |             | MCDB   | 9.1   | 8.7   | 9.4   | 10.1  | 11.3  | 11.8  | 10.9  | 11.0  | 11.1  | 10.7  | 9.8   | 8.7   |
|                                                                        | 5%          | WB     | 4.1   | 4.7   | 5.2   | 6.0   | 6.8   | 7.3   | 7.8   | 8.2   | 8.1   | 7.6   | 6.1   | 5.0   |
|                                                                        |             | MCDB   | 7.9   | 8.1   | 8.5   | 9.3   | 10.5  | 11.3  | 10.6  | 10.6  | 10.6  | 10.2  | 9.3   | 7.9   |
|                                                                        | 10%         | WB     | 3.0   | 3.7   | 4.2   | 5.2   | 6.0   | 6.5   | 7.2   | 7.6   | 7.5   | 6.8   | 5.2   | 3.9   |
|                                                                        |             | MCDB   | 8.0   | 7.3   | 7.4   | 8.6   | 9.8   | 10.7  | 10.3  | 10.3  | 10.3  | 9.7   | 8.4   | 7.0   |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 7.9   | 7.5   | 7.7   | 8.0   | 8.0   | 8.2   | 8.1   | 8.1   | 7.6   | 7.7   | 7.4   | 7.2   |
|                                                                        | 5% DB       | MCDBR  | 8.9   | 8.9   | 9.3   | 9.0   | 9.2   | 9.2   | 9.2   | 9.5   | 8.8   | 8.8   | 8.4   | 8.6   |
|                                                                        |             | MCWBR  | 6.9   | 7.7   | 8.3   | 8.6   | 8.7   | 7.9   | 7.8   | 8.2   | 8.6   | 8.0   | 7.7   | 6.9   |
|                                                                        | 5% WB       | MCDBR  | 7.9   | 7.5   | 8.0   | 7.6   | 7.9   | 7.7   | 7.3   | 7.3   | 7.3   | 7.3   | 7.5   | 6.8   |
|                                                                        |             | MCWBR  | 8.8   | 8.3   | 9.2   | 9.3   | 9.6   | 9.7   | 9.7   | 8.4   | 9.2   | 9.0   | 8.8   | 8.5   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.228 | 0.230 | 0.236 | 0.250 | 0.249 | 0.243 | 0.244 | 0.241 | 0.240 | 0.239 | 0.235 | 0.228 |
|                                                                        | taud        |        | 2.896 | 2.895 | 2.864 | 2.749 | 2.797 | 2.827 | 2.834 | 2.876 | 2.885 | 2.905 | 2.902 | 2.922 |
|                                                                        | Ebn at Noon |        | 1080  | 1086  | 1078  | 1053  | 1041  | 1038  | 1037  | 1049  | 1058  | 1062  | 1063  | 1072  |
|                                                                        | Edh at Noon |        | 67    | 71    | 76    | 86    | 81    | 78    | 78    | 75    | 73    | 69    | 66    | 63    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 4.50  | 4.81  | 5.13  | 5.55  | 5.73  | 5.97  | 6.18  | 6.03  | 5.51  | 4.95  | 4.34  | 4.09  |
|                                                                        | RadStd      |        | 0.26  | 0.35  | 0.57  | 0.38  | 0.40  | 0.21  | 0.32  | 0.32  | 0.24  | 0.21  | 0.24  | 0.40  |

## Historical Trends

|                             | DBAvg | Heating |     | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|-----|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  |     | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e) | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           | N/A   | N/A     | N/A | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (6 neighbors) | N/A   | N/A     | N/A | N/A     | N/A   | N/A   | N/A         | N/A     | +125    | +125    |

Nomenclature: See separate page